

Module 07 — Behavioral Patterns II: Communication

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Week 9 · Prerequisites: m06 (Strategy as lambdas, Command with undo), m05 (Facade), m03 (records with compact constructors) · Estimated study time: 6 h

```
Run          every          example          without          a          build:
java modules/m07-behavioral-communication/src/main/java/io/github/aliturgutbozkurt/
patterns/m07/examples/<path>/<Demo>.java
(JDK 27)
```

1 Learning outcomes

By the end of this module you can:

1. **Implement** Observer in its classic form (subject + observer interface) and its modern form (functional listeners returning an unsubscribe handle), and **explain** the lapsed-listener leak, notification order and re-entrancy pitfalls.
2. **Use** `java.util.concurrent.Flow` with `SubmissionPublisher`: **explain** `demand(request(n))`, back-pressure, buffering and dropping, and the `onSubscribe` → `onNext*` → `(onComplete | onError)` protocol — and **test** it deterministically.
3. **Implement** a Mediator that removes many-to-many references between colleagues, and **compare** it with an event bus (Observer) and with a Facade (m05).
4. **Implement** Chain of Responsibility both as linked handler objects and as composed functions (middleware), and **decide** between "first handler wins", "every handler runs" and "fail fast vs. collect all errors".
5. **Implement** Memento with an opaque classic memento and with record snapshots kept in a sequenced-collection history, and **compare** Memento-based undo with Command-based undo (m06).

2 Motivation

m06 put *algorithms* into objects. This module is about how objects *talk to each other* without knowing each other. A price changes and three screens must update; four aircraft want one runway; a request must pass authentication, logging and error handling before it reaches the code that answers it; a user wants to undo the last five edits. Wiring every sender directly to every receiver creates a tangle that nobody can change. The four patterns of this week cut those wires in four different ways: **announce** (Observer), **coordinate** (Mediator), **pass along** (Chain of Responsibility) and **remember** (Memento). The capstone starts this week — its domain events and request pipeline reuse two of these ideas directly.

3 Observer

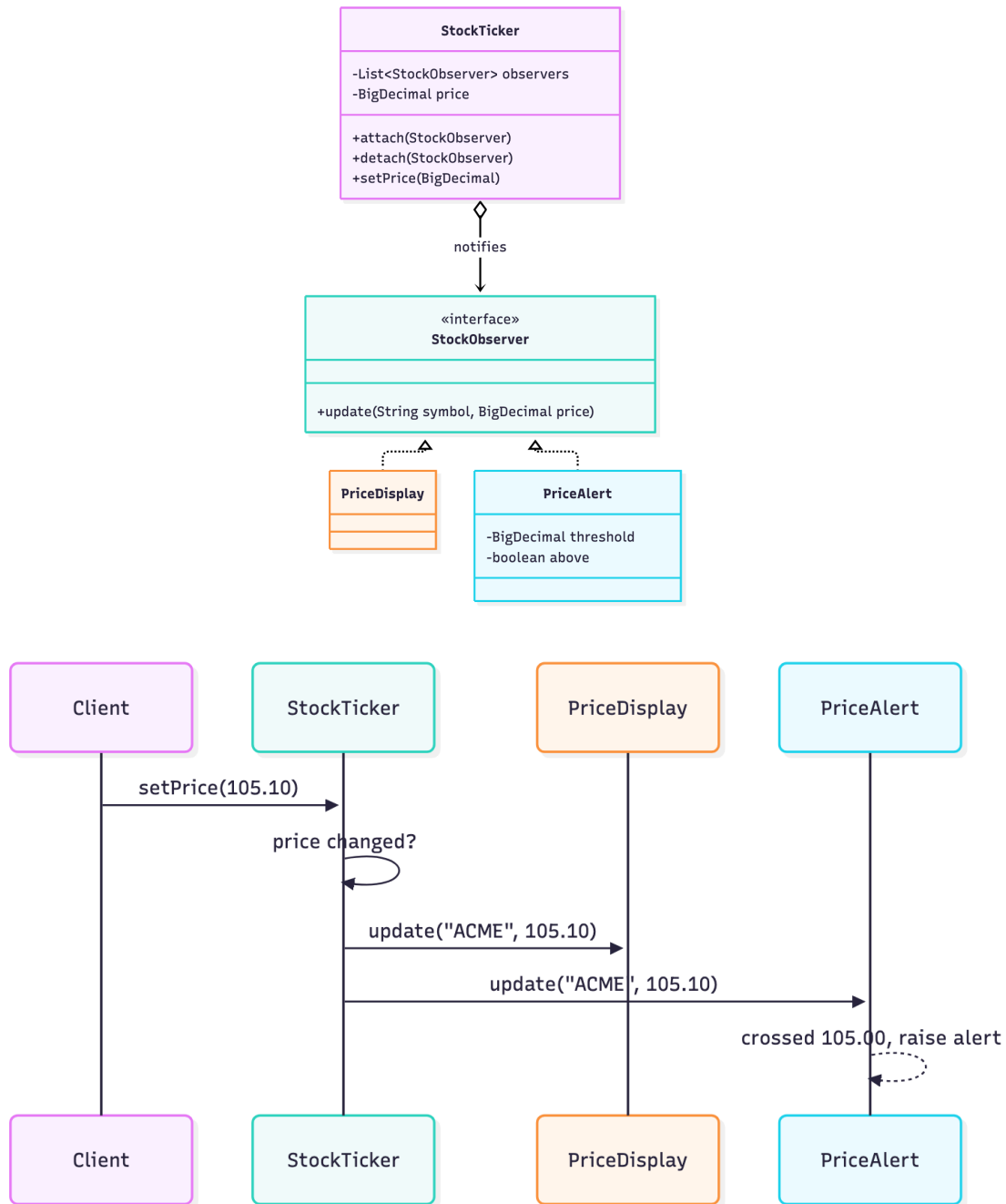
3.1 Problem

A stock ticker's price changes. A price display, an alert and — next month — a chart and a trading bot must react. If the ticker calls `display.show(...)`, `alert.check(...)` and `chart.plot(...)` itself, every new screen means editing the ticker, and the ticker cannot be reused without all of them.

3.2 Intent

Define a one-to-many dependency so that when one object (the **subject**) changes state, all its dependents (**observers**) are notified automatically — without the subject knowing their concrete classes.

3.3 Structure



3.4 Classic Java

The subject knows only the `StockObserver` interface. This is the **push** model: the subject sends the data with the notification (the **pull** model would pass the subject and let observers ask for what they need):

```
// file: examples/observer/classic/StockObserver.java
@FunctionalInterface
public interface StockObserver {

    /** Called after the price of {@code symbol} changed. */
    void update(String symbol, BigDecimal price);
}
```

Two details make the subject robust: no notification when nothing changed, and iteration over a **snapshot**, so an observer may detach itself while being notified without a `ConcurrentModificationException`:

```
// file: examples/observer/classic/StockTicker.java
public void setPrice(BigDecimal newPrice) {
    Objects.requireNonNull(newPrice, "newPrice");
    if (price.compareTo(newPrice) == 0) {
        return;
    }
    price = newPrice;
    notifyObservers();
}

private void notifyObservers() {
    // Iterate over a snapshot: an observer may attach or detach (itself) while
    // being notified.
    for (StockObserver observer : List.copyOf(observers)) {
        observer.update(symbol, price);
    }
}
```

Observers may keep their own state. The alert fires only when the price *crosses* the threshold, not on every update above it:

```
// file: examples/observer/classic/PriceAlert.java
@Override
public void update(String symbol, BigDecimal price) {
    boolean nowAbove = price.compareTo(threshold) >= 0;
    if (nowAbove != above) {
        String direction = nowAbove ? " rose above " : " fell below ";
        alerts.accept("ALERT " + symbol + direction + threshold.toPlainString() + ": "
            + price.toPlainString());
    }
    above = nowAbove;
}
```

```
ACME 101.50
ACME 104.20
ACME 105.10
ALERT ACME rose above 105.00: 105.10
ACME 106.00
-- display detached --
ALERT ACME fell below 105.00: 99.80
```

The JDK once shipped this exact shape as `java.util.Observable/Observer`. It is `@Deprecated(since = "9")`: `Observable` is a class (you must extend it), its `setChanged()` is protected, events are untyped `Object`s and the notification order is unspecified.

3.5 Modern Java 27

Functional listeners and a subscription handle. An observer with one method is just a `Consumer<PriceChange>`. Subscribing returns a `Subscription` whose `close()` throws no checked exception and is idempotent:

```
// file: examples/observer/modern/Subscription.java
@FunctionalInterface
public interface Subscription extends AutoCloseable {

    /** Stops delivery to the listener; calling it again does nothing. */
    @Override
    void close();
}
```

```
// file: examples/observer/modern/Ticker.java
public Subscription onPriceChange(Consumer<? super PriceChange> listener) {
    var registration = new Registration(Objects.requireNonNull(listener,
"listener"));
    registrations.add(registration);
    return () -> registrations.remove(registration); // removing twice is harmless:
idempotent
}
// ...
for (Registration registration : registrations) {
    try {
        registration.listener.accept(change);
    } catch (RuntimeException e) {
        errorHandler.accept(e); // not swallowed: reported, and the remaining
listeners still run
    }
}
```

Three decisions are visible here. Each subscription gets its own `Registration` object, so closing one subscription never removes a second registration of the *same* lambda. `registrations` is a `CopyOnWriteArrayList`: delivery iterates over a snapshot, so a listener added during delivery receives only later events. And one failing listener is reported to an injected error handler instead of silencing everyone after it.

The handle fixes the **lapsed-listener leak**: a subject keeps every listener — and everything the listener refers to — alive until it is removed. With `try-with-resources` the removal cannot be forgotten; the resource variable is the unnamed `_` because the body never uses it:

```
// file: examples/observer/ModernTickerDemo.java
try (var _ = ticker.onPriceChange(ModernTickerDemo::chart)) {
```

```
chart: ACME 100.00 -> 101.50 (+1.50)
error handler: ticker feed rejected ACME 101.50
log: ACME 101.50
chart: ACME 101.50 -> 99.00 (-2.50)
error handler: ticker feed rejected ACME 99.00
log: ACME 99.00
-- chart closed (try-with-resources) --
error handler: ticker feed rejected ACME 98.25
log: ACME 98.25
```

The JDK's own observer for beans. `java.beans.PropertyChangeSupport` (used by Swing and IDE tooling) delivers the property name with the old and the new value, and fires nothing when an equal value is set:

```
// file: examples/observer/beans/Thermostat.java
    int oldTarget = target;
    target = newTarget;
    changes.firePropertyChange("target", oldTarget, newTarget);
```

```
// file: examples/observer/ThermostatDemo.java
    thermostat.addPropertyChangeListener(event -> print("app:    ",
event)); // every property
    thermostat.addPropertyChangeListener("target", event -> print("display: ",
event)); // one property
```

```
app:    target 20 -> 22
display: target 20 -> 22
app:    mode OFF -> HEAT
(setting target 22 again fires nothing)
app:    target 22 -> 19
display: target 22 -> 19
```

A typed event bus. In PatternShop, the capstone's online shop, publishers and handlers should not even know the subject they share. An event bus is an Observer whose events form a `sealed` hierarchy of records (`OrderPlaced`, `PaymentFailed`, `OrderShipped`); handlers subscribe by type, and a handler for `ShopEvent` sees everything — with an exhaustive `switch`, so a new event type is a compile error, not a silently ignored event:

```
// file: examples/observer/eventbus/AuditLog.java
@Override
public void accept(ShopEvent event) {
    out.accept(switch (event) {
        case OrderPlaced(var id, var customer, var total) -> "order " + id + "
placed by " + customer
            + ", total " + total.toPlainString();
        case PaymentFailed(var id, var reason) -> "payment failed for " + id + ": "
+ reason;
        case OrderShipped(var id, var tracking) -> "order " + id + " shipped,
tracking " + tracking;
    });
}
```

What happens when a handler publishes an event while the bus is still delivering another one? A naive bus recurses: the new event overtakes the one being delivered, and later handlers see them in the wrong order. This bus queues it:

```
// file: examples/observer/eventbus/EventBus.java
public void publish(ShopEvent event) {
    pending.addLast(Objects.requireNonNull(event, "event"));
    if (dispatching) {
        return; // called from inside a handler: the running loop below will deliver
it next
    }
    dispatching = true;
    try {
        while (!pending.isEmpty()) {
            dispatch(pending.removeFirst());
        }
    }
```

```

    }
    } finally {
        dispatching = false;
        pending.clear(); // non-empty only if a handler threw: drop what it queued
    }
}

```

```

// file: examples/observer/EventBusDemo.java
bus.subscribe(OrderPlaced.class, placed -> {
    System.out.println("warehouse: shipping " + placed.orderId());
    bus.publish(new OrderShipped(placed.orderId(), "TRK-" +
placed.orderId())); // queued, not recursive
});

```

The audit log still sees "placed" before "shipped", and an event nobody subscribed to is kept as a *dead event* instead of vanishing:

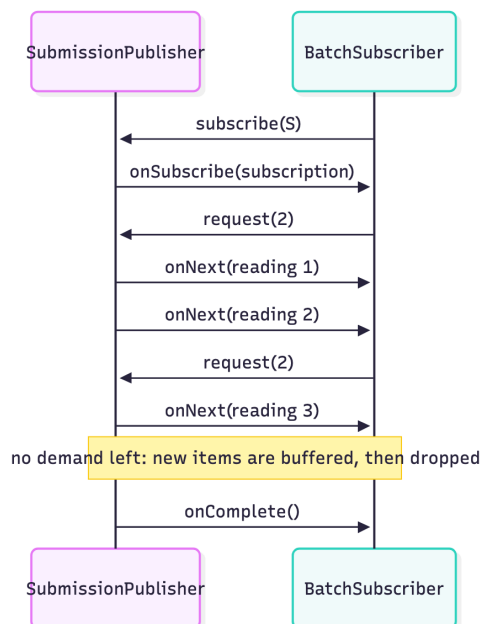
```

warehouse: shipping A-1
audit:    order A-1 placed by ada, total 42.00
audit:    order A-1 shipped, tracking TRK-A-1
audit:    payment failed for A-2: card declined
mailer:   e-mail sent: payment for A-2 failed (card declined)
dead events: [OrderPlaced[orderId=A-0, customer=alan, total=10.00]]

```

3.6 Back-pressure with Flow

A listener is *pushed* every event at the speed of the subject. If a sensor produces readings faster than a subscriber can store them, something must give: an unbounded queue grows until memory runs out. Reactive Streams — in the JDK as `java.util.concurrent.Flow` since Java 9 — adds **demand**: a subscriber receives at most as many items as it has requested with `request(n)`. That is **back-pressure**.



The protocol is always `onSubscribe` → `onNext*` → (`onComplete` | `onError`). A `BatchSubscriber` controls its own pace: one batch on subscribe, the next one only when the current batch has arrived:

```
// file: examples/observer/flow/BatchSubscriber.java
@Override
public void onNext(T item) {
    received.add(item);
    signals.add("onNext " + item);
    remainingInBatch--;
    if (remainingInBatch == 0) {
        requestBatch();
    }
}
// ...
private void requestBatch() {
    remainingInBatch = batchSize;
    requestCount++;
    signals.add("request(" + batchSize + ")"); // logged first: with a caller-runs
    executor.request() delivers at once
    subscription.request(batchSize);
}
```

The publisher is the JDK's `SubmissionPublisher`. It has two ways to hand over an item. `submit` **blocks** the publishing thread while a subscriber's buffer is full; `offer` never blocks — it calls a drop handler instead. The feed offers and counts what it had to drop:

```
// file: examples/observer/flow/TemperatureFeed.java
public void publish(Reading reading) {
    publisher.offer(Objects.requireNonNull(reading, "reading"), (subscriber, item) -
> {
        dropped.incrementAndGet();
        return false; // do not retry
    });
}
```

Testing asynchronous code deterministically. By default `SubmissionPublisher` delivers on `ForkJoinPool.commonPool()`, so output order depends on thread timing. Pass a *caller-runs* executor, `Runnable::run`, and every `onSubscribe`, `onNext` and `onComplete` happens synchronously inside `subscribe`, `publish`, `request` and `close`. No sleeps, no latches:

```
// file: examples/observer/FlowDemo.java
var batches = new BatchSubscriber<Reading>(2);
try (var feed = new TemperatureFeed(Runnable::run, 4)) {
    feed.subscribe(batches);
    publishAll(feed);
}
```

With a fast subscriber and a slow one that has not requested anything, a buffer of 2 holds two readings for the slow one and the other three are dropped — for it only. Buffered items are still delivered before `onComplete`:

```
== 1. demand in batches of 2
onSubscribe
request(2)
onNext greenhouse#1 20.0C
onNext greenhouse#2 22.5C
request(2)
```

```

onNext greenhouse#3 25.0C
onNext greenhouse#4 17.5C
request(2)
onNext greenhouse#5 20.0C
onComplete
== 2. a slow subscriber and a buffer of 2
fast received 5, slow received 0, dropped 3
slow: onSubscribe
slow: request(5)
slow: onNext greenhouse#1 20.0C
slow: onNext greenhouse#2 22.5C
slow: onComplete
== 3. processor stage C -> F
greenhouse#1 68.0F
greenhouse#2 72.5F
greenhouse#3 77.0F
greenhouse#4 63.5F
greenhouse#5 68.0F
== 4. the sensor fails
onSubscribe
request(2)
onNext greenhouse#1 20.0C
onError IllegalStateException: sensor offline

```

A **processor** is a subscriber and a publisher at once — a pipeline stage. It requests one item upstream at a time and uses `submit`, whose blocking is exactly how back-pressure travels upstream:

```

// file: examples/observer/flow/CelsiusToFahrenheit.java
@Override
public void onNext(Reading reading) {
    submit(new FahrenheitReading(reading.sensor(), reading.sequence(),
reading.celsius() * 9 / 5 + 32));
    upstream.request(1);
}

```

(With a caller-runs executor, `submit` to a subscriber that never requests would block its own thread forever — so the demos use `offer` wherever demand is bounded, and exactly one test runs on the real pool, waiting on the future returned by `consume(...)` with a 5-second timeout.)

3.7 Real-world usage

`java.beans.PropertyChangeSupport`, Swing/AWT listeners (`ActionListener`), JavaFX properties, `java.util.concurrent.Flow` and `SubmissionPublisher`, `java.net.http.HttpClient` (body publishers/subscribers are `Flow` types), Spring's `ApplicationEventPublisher` and `@EventListener`, Guava's `EventBus`, and every reactive library (Reactor, RxJava) built on the Reactive Streams interfaces.

3.8 Pitfalls and when NOT to use it

- **Lapsed listeners** leak memory: always return and use an unsubscribe handle.
- **Order and re-entrancy**: define the notification order, iterate over a snapshot, queue events published from inside a handler.
- **One failing listener** must not break the others — catch, report, continue (never swallow silently).
- **Cascades**: observers that change other subjects can trigger long, hard-to-debug chains of updates.
- A single, known receiver does not need `Observer` — call it directly.
- `Flow` is for streams with a real producer/consumer speed mismatch; for a handful of UI events it is overkill.

3.9 Related patterns

Mediator centralises the communication that Observer distributes (an event bus sits in between). **Command** (m06) objects are often what an event bus carries. **Chain of Responsibility** asks receivers one by one until one handles a request; Observer tells all of them.

4 Mediator

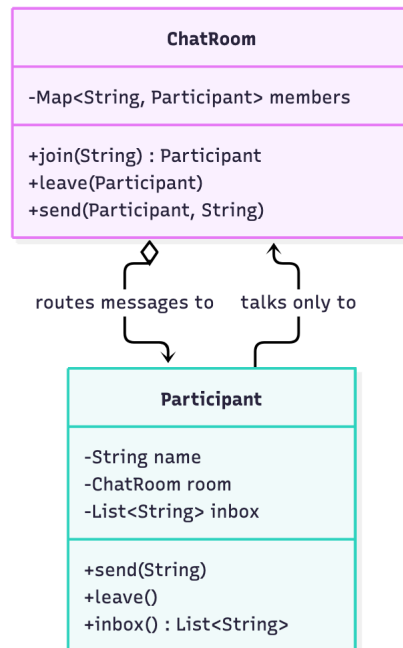
4.1 Problem

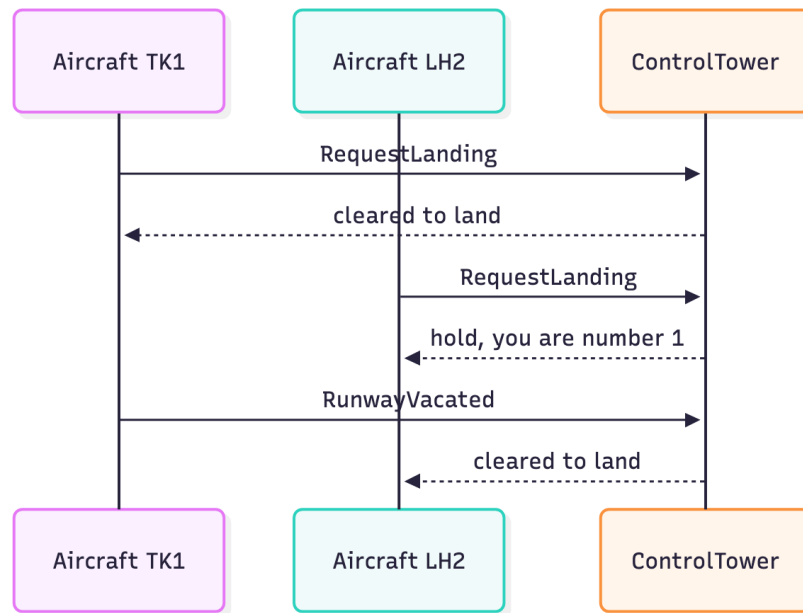
Four aircraft share one runway. If each aircraft asks every other aircraft whether the runway is free, every aircraft must know every other one: $n \cdot (n-1)$ connections, and the queueing rule is copied into every aircraft. The same happens in a dialog where a checkbox enables a text field, which enables a button.

4.2 Intent

Define an object that **encapsulates how a set of objects interact**. Colleagues refer only to the mediator, never to each other, so their interaction can be changed in one place.

4.3 Structure





4.4 Classic Java

Participants hold a reference to the room and nothing else; the room owns the routing rules — broadcast, direct @name messages, and a notice back to the sender when the addressee is unknown:

```
// file: examples/mediator/chat/ChatRoom.java
public void send(Participant sender, String text) {
    Objects.requireNonNull(text, "text");
    if (members.get(sender.name()) != sender) {
        throw new IllegalStateException(sender.name() + " is not in " + name);
    }
    if (text.startsWith("@")) {
        int space = text.indexOf(' ');
        String addressee = space < 0 ? text.substring(1) : text.substring(1, space);
        String body = space < 0 ? "" : text.substring(space + 1);
        Participant recipient = members.get(addressee);
        if (recipient == null) {
            sender.receive(name + ": nobody called '" + addressee + "' is here");
        } else {
            recipient.receive("(private) " + sender.name() + ": " + body);
        }
        return;
    }
    for (Participant member : members.values()) {
        if (member != sender) {
            member.receive(sender.name() + ": " + text);
        }
    }
}
```

```
// file: examples/mediator/chat/Participant.java
/** Sends through the room; the room decides who receives it. */
public void send(String text) {
    room.send(this, text);
}
```

A test checks by reflection that `Participant` declares no field of type `Participant` — the colleagues really are decoupled:

```
ada []
bob [ada: hello everyone, ada: cem left early]
cem [ada: hello everyone, (private) bob: lunch at noon?, #patterns: nobody called 'dave'
is here]
```

4.5 Modern Java 27

In the classic form colleagues call `mediator.notify(this, "someEvent")` with a string that the mediator decodes. With a sealed interface of request records the mediator handles every possible message in one **exhaustive** switch with record patterns — add a request type and the compiler shows where it must be handled:

```
// file: examples/mediator/atc/TowerRequest.java
public sealed interface TowerRequest permits RequestLanding, RequestTakeoff,
DeclareEmergency, RunwayVacated {
```

```
// file: examples/mediator/atc/ControlTower.java
switch (request) {
    case RequestLanding(var aircraft) -> useRunwayOrWait(aircraft, "cleared to
land");
    case RequestTakeoff(var aircraft) -> useRunwayOrWait(aircraft, "cleared for
takeoff");
    case DeclareEmergency(var aircraft) -> {
        queue.removeIf(waiting -> waiting.aircraft() == aircraft);
        if (onRunway == null) {
            clear(aircraft, "cleared for emergency landing");
        } else {
            queue.addFirst(new Waiting(aircraft, "cleared for emergency
landing")); // jumps the queue
            transmit(aircraft, "emergency acknowledged, you are number 1");
        }
    }
    case RunwayVacated(var aircraft) -> {
        if (onRunway != aircraft) {
            throw new IllegalStateException(aircraft + " is not on the runway");
        }
        onRunway = null;
        Waiting next = queue.pollFirst();
        if (next != null) {
            clear(next.aircraft(), next.clearance());
        }
    }
}
```

The tower holds the shared resource (the runway) and the waiting queue; at most one aircraft is ever cleared:

```
TK1 -> tower: RequestLanding
tower -> TK1: cleared to land
LH2 -> tower: RequestLanding
tower -> LH2: hold, you are number 1
BA3 -> tower: RequestTakeoff
```

```

tower -> BA3: hold, you are number 2
AF4 -> tower: DeclareEmergency
tower -> AF4: emergency acknowledged, you are number 1
TK1 -> tower: RunwayVacated
tower -> AF4: cleared for emergency landing
AF4 -> tower: RunwayVacated
tower -> LH2: cleared to land
LH2 -> tower: RunwayVacated
tower -> BA3: cleared for takeoff
BA3 -> tower: RunwayVacated

```

4.6 GUI forms

Mediator was born in GUI toolkits. In a sign-up dialog, "business account" enables the company field, and "Submit" is enabled only when everything is valid. Each widget only calls `form.changed(this)`; every dependency lives in one method (the example is headless — no Swing needed):

```

// file: examples/mediator/form/SignUpForm.java
void changed(Widget source) {
    if (source == submit) {
        submissions.add(email.text() + (business.isChecked() ? " (business: " +
company.text() + ")" : " (personal)"));
        return;
    }
    if (source == business) {
        if (!business.isChecked()) {
            company.clear();
        }
        company.setEnabled(business.isChecked());
    }
    submit.setEnabled(isComplete());
}

```

```

new form          -> submit disabled, company disabled
e-mail + password -> submit disabled, company disabled
terms accepted    -> submit enabled, company disabled
business account  -> submit disabled, company enabled ''
company named     -> submit enabled, company enabled 'Analytical Engines'
personal again    -> submit enabled, company disabled
submitted: [ada@example.com (personal)]

```

4.7 Real-world usage

`javax.swing.ButtonGroup` (selecting one radio button deselects the others — the buttons never reference each other), dialog controllers in desktop toolkits, `java.util.concurrent.Exchanger` (two threads meet through it), chat servers and message brokers, air-traffic control, and "controller"/"coordinator" objects in MVC frameworks.

4.8 Pitfalls and when NOT to use it

- **The god mediator:** all logic drifts into the mediator until it is the hardest class in the system. Keep it about *coordination*; business rules belong in the colleagues or in services.
- Two colleagues that simply call each other do not need a mediator.
- A mediator is a single point of failure and, if shared between threads, a bottleneck.

4.9 Related patterns

Observer: colleagues often notify the mediator through events; an event bus is a mediator that only broadcasts. **Facade** (m05) also sits in front of several objects, but communication is one-way — clients call the facade, the subsystem does not know it — whereas a mediator is known by and talks back to its colleagues.

5 Chain of Responsibility

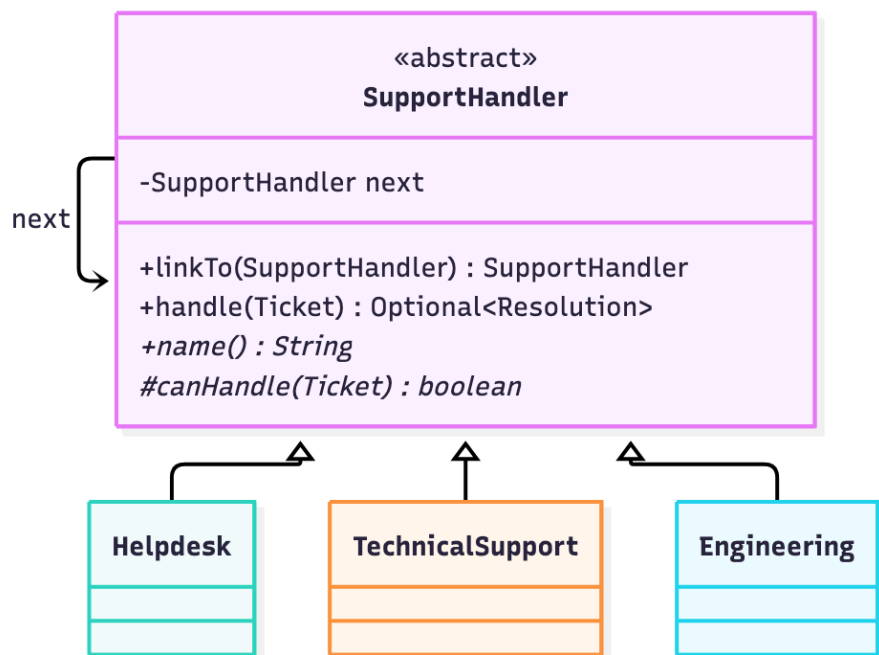
5.1 Problem

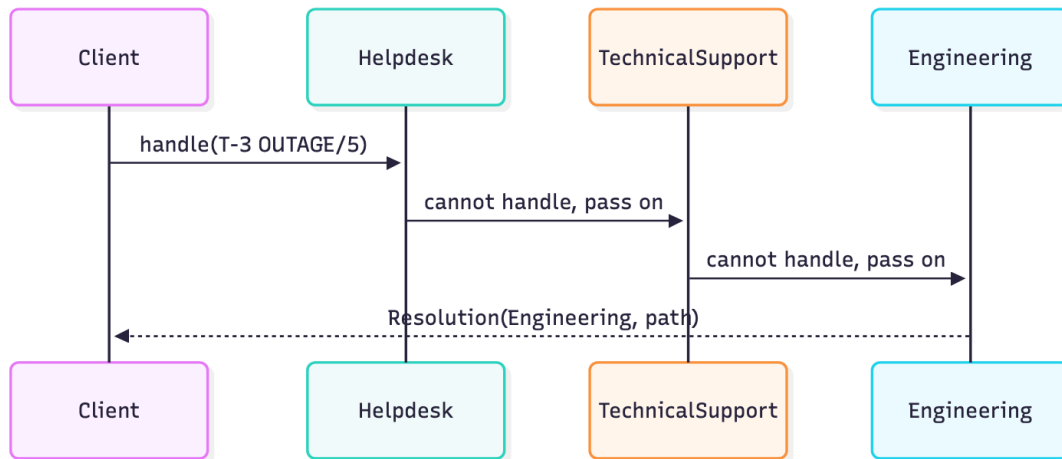
A support ticket may be solved by the helpdesk, by technical support or only by engineering. The customer should not need to know who handles what, and the escalation rules change every quarter. One method full of nested `if`s, naming every level, would change with every reorganisation.

5.2 Intent

Avoid coupling the sender of a request to its receiver by giving **more than one object a chance to handle it**. Chain the receivers and pass the request along the chain until an object handles it.

5.3 Structure





5.4 Classic Java

Each handler either resolves the ticket or forwards it to its successor; the path is recorded on the way:

```
// file: examples/chain/support/classic/SupportHandler.java
private Optional<Resolution> handle(Ticket ticket, List<String> path) {
    path.add(name());
    if (canHandle(ticket)) {
        return Optional.of(new Resolution(ticket.id(), name(), path));
    }
    return next == null ? Optional.empty() : next.handle(ticket, path);
}
```

```
// file: examples/chain/support/classic/Helpdesk.java
@Override
protected boolean canHandle(Ticket ticket) {
    return (ticket.topic() == Topic.PASSWORD || ticket.topic() == Topic.BILLING) &&
    ticket.severity() <= 2;
}
```

What happens when nobody handles the request? The chain must say so explicitly — here with `Optional.empty()` — instead of silently dropping it.

5.5 Modern Java 27

A handler that returns `Optional<Resolution>` — "a result, or not me" — is a function, and two such functions compose with `Optional.or()`:

```
// file: examples/chain/support/modern/TicketHandler.java
default TicketHandler orElse(TicketHandler next) {
    Objects.requireNonNull(next, "next");
    TicketHandler first = this;
    List<String> levels = Stream.concat(first.levels().stream(),
next.levels().stream()).toList();
    return new TicketHandler() {
        @Override
        public Optional<Resolution> handle(Ticket ticket) {
            return first.handle(ticket)
                .or(() -> next.handle(ticket).map(resolution ->
```

```
resolution.escalatedFrom(first.levels())));
}
```

```
// file: examples/chain/SupportEscalationDemo.java
TicketHandler modern = SupportLevels.helpdesk()
    .orElse(SupportLevels.technicalSupport())
    .orElse(SupportLevels.engineering());
```

Without the escalation path, `orElse` would be a one-liner: `ticket -> handle(ticket).or(() -> next.handle(ticket))`. The linked chain got the path for free (each object adds its name as it forwards); stateless functions must carry it explicitly, hence `levels()`. A parameterised test runs both chains over the same table of tickets and requires equal results; a call counter proves that handlers after the deciding one are never invoked:

```
classic chain (linked objects):
T-1 PASSWORD/1 -> Helpdesk via [Helpdesk]
T-2 BUG/2 -> Technical support via [Helpdesk, Technical support]
T-3 OUTAGE/5 -> Engineering via [Helpdesk, Technical support, Engineering]
T-4 LEGAL/1 -> unresolved (end of chain)
modern chain (composed functions):
T-1 PASSWORD/1 -> Helpdesk via [Helpdesk]
T-2 BUG/2 -> Technical support via [Helpdesk, Technical support]
T-3 OUTAGE/5 -> Engineering via [Helpdesk, Technical support, Engineering]
T-4 LEGAL/1 -> unresolved (end of chain)
same resolutions: true
```

5.6 Middleware

In a server every link may act **before and after** its successor, or stop the request. That is how servlet `Filter/FilterChain.doFilter` and the JDK's `com.sun.net.httpserver.Filter.Chain` work. A middleware is a function from handler to handler:

```
// file: examples/chain/middleware/Middleware.java
@FunctionalInterface
public interface Middleware {

    Handler wrap(Handler next);
}
```

```
// file: examples/chain/middleware/Pipeline.java
public static Handler of(List<Middleware> middlewares, Handler endpoint) {
    Handler handler = Objects.requireNonNull(endpoint, "endpoint");
    for (Middleware middleware : middlewares.reversed()) { // wrap from the inside
        handler = middleware.wrap(handler);
    }
    return handler;
}
```

The first middleware in the list is the outermost one: requests pass the list in declared order, responses come back in reverse. Authentication short-circuits — the endpoint is never called without a valid token:

```
// file: examples/chain/middleware/Middlewares.java
public static Middleware authentication(Set<String> tokens) {
    Set<String> known = Set.copyOf(tokens);
    return next -> request -> request.header("Authorization")
        .filter(value -> value.startsWith("Bearer ") &&
known.contains(value.substring("Bearer ".length())))
        .map(_ -> next.handle(request))
        .orElseGet(() -> new Response(401, "unauthorized"));
}
```

```
// file: examples/chain/MiddlewareDemo.java
Handler server = Pipeline.of(List.of(
    Middlewares.logging(log),
    Middlewares.errorBoundary(),
    Middlewares.authentication(Set.of("s3cret")),
    Middlewares.requestId(() -> "req-" + ids.incrementAndGet()), endpoint);
```

```
-> GET /orders
<- 200 GET /orders
    200 orders for req-1
-> GET /orders
<- 401 GET /orders
    401 unauthorized
-> GET /crash
<- 500 GET /crash
    500 internal error: database down
```

Order matters: `logging` is outermost, so it also logs the 401 and the 500; `errorBoundary` sits outside `authentication`, so it would also catch a failing token check.

5.7 Validation chains

Validation is a chain in which the policy is a choice. **Collect all**: every link runs and all errors are reported (good for forms). **Fail fast**: the first failure stops the chain (good when later checks are expensive or depend on earlier ones):

```
// file: examples/chain/validation/Validator.java
default Validator<T> and(Validator<? super T> next) {
    Objects.requireNonNull(next, "next");
    return value -> ValidationResult.merge(validate(value), next.validate(value));
}

/** Fail fast: runs {@code next} only if this validator passed. */
default Validator<T> andThen(Validator<? super T> next) {
    Objects.requireNonNull(next, "next");
    return value -> switch (validate(value)) {
        case Valid _ -> next.validate(value);
        case Invalid invalid -> invalid;
    };
}
```

```
SignUp[email=ada.example.com, password=short, age=16]
  collect-all: Invalid[errors=[email must contain @, password must have at least 8
characters, password must contain a digit, age must be at least 18]]
  fail-fast:   Invalid[errors=[email must contain @]]
```

```
SignUp[email=ada@example.com, password=s3cret-pass, age=36]
  collect-all: Valid[]
  fail-fast:   Valid[]
```

5.8 Real-world usage

Servlet `Filter` and `FilterChain`, `com.sun.net.httpserver.Filter`, Spring Security's filter chain, logging frameworks (a log event travels up the logger hierarchy to its appenders), `java.util.logging.Logger` parent handlers, exception handling itself (a stack of `catch` blocks, innermost first), and event bubbling in GUI toolkits.

5.9 Pitfalls and when NOT to use it

- **Nobody handles it:** decide what the end of the chain means — a default handler, an explicit "unresolved" result, or an exception — and test it.
- **Order bugs:** a chain is only as correct as its order (authentication after the endpoint is useless).
- **Hard to debug:** a request may silently pass through ten links; record the path (as `Resolution` does) or log it.
- If exactly one fixed receiver handles every request, a chain only adds indirection.

5.10 Related patterns

Decorator (m04) has the same "wrap the next one" structure as middleware, but a decorator always delegates and adds behaviour, while a chain link may decide not to. **Composite** (m05): a request often travels up a composite's parent chain. **Command** (m06) objects are what chains typically carry.

6 Memento

6.1 Problem

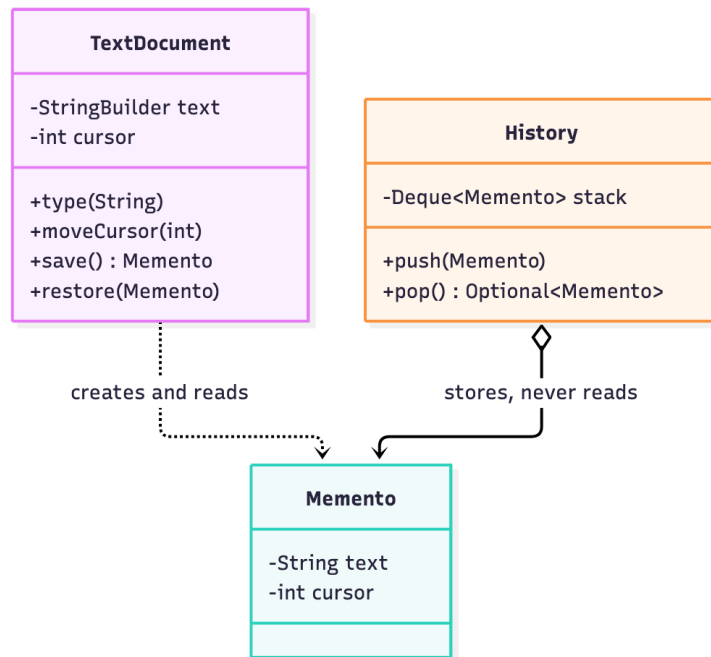
A text editor needs undo. The editor's state — text, cursor, selection — is private, as it should be. Letting an undo manager read and write those fields would break encapsulation; copying the whole editor would copy far too much.

6.2 Intent

Without violating encapsulation, **capture and externalise an object's internal state** so that the object can be restored to this state later.

The roles are the **originator** (the object whose state is saved), the **memento** (the saved state) and the **caretaker** (keeps mementos, never looks inside).

6.3 Structure



6.4 Classic Java

The memento is a nested class with private fields, a private constructor and no accessors. Only the enclosing originator can create or read it; the caretaker can store it and hand it back, nothing else:

```
// file: examples/memento/classic/TextDocument.java
public static final class Memento {
    private final String text;
    private final int cursor;

    private Memento(String text, int cursor) {
        this.text = text;
        this.cursor = cursor;
    }
}
// ...
public Memento save() {
    return new Memento(text.toString(), cursor);
}

public void restore(Memento memento) {
    Objects.requireNonNull(memento, "memento");
    text.setLength(0);
    text.append(memento.text);
    cursor = memento.cursor;
}
}
```

```
saved: Hello|
saved: Hello, world|
edited: >> |Hello, world
undo: Hello, world|
undo: Hello|
nothing to undo
```

6.5 Modern Java 27

Record snapshots. A record's components are readable by everyone, so a record memento is *not* opaque. It trades opacity for something else: it is immutable, so it can be shared, compared with `equals` and validated once — in the compact constructor — however it was created:

```
// file: examples/memento/editor/EditorSnapshot.java
public record EditorSnapshot(String text, int cursor, int selectionStart, int
selectionEnd) {

    public EditorSnapshot {
        Objects.requireNonNull(text, "text");
        if (cursor < 0 || cursor > text.length()) {
            throw new IllegalArgumentException("cursor " + cursor + " outside 0.." +
text.length());
        }
    }
}
```

A bounded history with sequenced collections. The undo stack keeps the newest snapshot *last* (addLast, removeLast, getLast), forgets the oldest one with removeFirst, and reversed() gives a newest-first view — the method names say exactly what happens (JEP 431):

```
// file: examples/memento/editor/UndoHistory.java
public Optional<EditorSnapshot> undo(EditorSnapshot current) {
    if (undo.isEmpty()) {
        return Optional.empty();
    }
    redo.addLast(current);
    return Optional.of(undo.removeLast());
}
// ...
public List<EditorSnapshot> history() {
    return List.copyOf(undo.reversed());
}

private void pushUndo(EditorSnapshot snapshot) {
    undo.addLast(snapshot);
    if (undo.size() > capacity) {
        undo.removeFirst(); // bounded: forget the oldest state
    }
}
}
```

A new edit clears the redo stack, and with capacity 3 only three steps can be undone:

```
select -> Hello [world]
type -> Hello Java|
undo true -> Hello [world]
undo true -> Hello world|
redo true -> Hello [world]
type -> Hello there|
redo false -> Hello there|
history (newest first, capacity 3): [Hello [world], Hello world|, Hello|]
```

Deep immutability. A record is only as immutable as its components. `GameState` copies the inventory with `List.copyOf` in the compact constructor, so picking up an item after saving can never change the saved game:

```
// file: examples/memento/game/GameState.java
Objects.requireNonNull(position, "position");
inventory = List.copyOf(inventory);
```

Save slots are a `SequencedMap` in *write* order. `putLast` moves a re-saved slot to the end (`put` would keep its old position), `lastEntry()` implements "Continue" and `pollFirstEntry()` evicts the least recently written slot:

```
// file: examples/memento/game/SaveSlots.java
public Optional<String> save(String slot, GameState state) {
    Objects.requireNonNull(slot, "slot");
    Objects.requireNonNull(state, "state");
    Optional<String> evicted = Optional.empty();
    if (!slots.containsKey(slot) && slots.size() == maxSlots) {
        evicted = Optional.of(slots.pollFirstEntry().getKey()); // least recently
written
    }
    slots.putLast(slot, state); // put() would keep an existing slot at its old
position
    return evicted;
}
// ...
public Optional<GameState> continueLatest() {
    return Optional.ofNullable(slots.lastEntry()).map(Map.Entry::getValue);
}
```

```
save castle:  level 1, health 100, at (3,4), inventory [sword]
save tower:   level 2, health 70, at (3,4), inventory [sword, key]
oops:         level 2, health 0, at (3,4), inventory [sword, key]
load castle:  level 1, health 100, at (3,4), inventory [sword]
slots:        [tower, castle]
continue:     level 1, health 100, at (5,5), inventory [sword]
save forest evicts tower
load tower:   no save slot 'tower'; known slots: [castle, cave, forest]
```

Memento-based vs. Command-based undo (m06). Command undo stores *operations* and their inverses — small, but every command needs a correct `undo()`. Memento undo stores *states* — no inverse logic, always correct, but each step costs a full snapshot. Editors often combine them: commands for small edits, periodic snapshots as checkpoints.

6.6 Real-world usage

Undo/redo in editors and drawing tools, game save slots and checkpoints, database transactions and savepoints (rollback restores a previous state), `java.io.Serializable` object snapshots, and immutable state stores in UI frameworks where every state is a snapshot and "time travel" means keeping the old ones.

6.7 Pitfalls and when NOT to use it

- **Memory:** every snapshot is a full copy — bound the history (capacity, eviction) and share immutable parts.
- **Shallow snapshots:** a mutable list inside a memento lets later edits leak into the "saved" state; use `List.copyOf` / records all the way down.
- **Opacity vs. convenience:** a record memento can be read by the caretaker — fine inside one module, not across a trust boundary.
- If the state is huge and changes are small, Command-based undo is cheaper.

6.8 Related patterns

Command (m06) is the alternative way to undo and often *uses* mementos to store what it needs to revert.

Prototype (m03) copies an object to create a new one; Memento copies its state to restore the same one.

Iterator (m06) may use a memento to remember a position.

7 Choosing a communication pattern

Situation	Use
One change, many interested parties that come and go	Observer (listeners + subscription handle)
Producer faster than consumer, items must not pile up	Flow with demand (<code>request(n)</code>), <code>offer</code> + drop policy
Publishers and handlers must not know each other at all	Event bus (typed, sealed events)
Many objects coordinating over a shared resource or rule	Mediator
A request that one of several handlers should take	Chain of Responsibility (first one wins)
Every link must see the request, before and after	Middleware pipeline
All checks must run and all errors be reported	Validation chain, collect-all (and)
Restore an earlier state without exposing internals	Memento (opaque or record snapshot)

8 Summary

Pattern	Use when	Avoid when	Java 27 shortcut
Observer	Unknown number of receivers react to changes	One known receiver	<code>Consumer</code> listeners + <code>AutoCloseable</code> subscription, <code>CopyOnWrite ArrayList</code>
Observer (Flow)	Stream with a speed mismatch	A few UI events	<code>SubmissionPublisher</code> , <code>Runnable::run</code> in tests
Mediator	Many-to-many interaction between colleagues	Two objects that just call each other	Sealed request records + exhaustive <code>switch</code>
Chain of Responsibility	Several possible handlers, sender must not choose	One fixed receiver	<code>Optional.or</code> , <code>Handler</code> → <code>Handler</code> middleware
Memento	Undo, checkpoints, save/load	Huge state, tiny changes (use <code>Command</code>)	<code>Records</code> + <code>List.copyOf</code> of, sequenced collections

9 Quiz

1. What is the difference between push and pull Observer, and which one does `StockObserver` use?
2. Why was `java.util.Observable` deprecated?
3. What is the lapsed-listener leak, and how does a `Subscription` used in try-with-resources prevent it?

4. A handler publishes a new event while the bus is still delivering another one. What goes wrong without a queue?
5. What does `request(n)` mean, and what does back-pressure protect?
6. When does `SubmissionPublisher.submit` block, and why do the demos use `offer` with a caller-runs executor?
7. How does a Mediator differ from an event bus and from a Facade (m05)?
8. In a middleware pipeline declared as `[logging, errorBoundary, authentication]`, in which order do the three run on the way in and on the way out — and does `logging` see a 401?
9. Why is the classic memento opaque, what does a record memento give up, and what does it gain?
10. Name one advantage of Memento-based undo and one of Command-based undo (m06).

Answers

1. Push sends the changed data with the notification; pull sends only "something changed" (or the subject) and the observer asks for what it needs. `StockObserver.update(symbol, price)` is push.
2. `Observable` is a class you must extend (no multiple inheritance), `setChanged()` is protected, events are untyped `Object`s, the notification order is unspecified and it is not serializable or thread-safe in a useful way.
3. A subject keeps a strong reference to every listener, so a forgotten listener (and everything it references) is never garbage-collected. The handle removes exactly that registration, and try-with-resources calls `close()` even on an exception.
4. Delivery becomes recursive: the new event reaches later handlers *before* the event that caused it, so they see "shipped" before "placed". Queuing delivers it after the current event finishes.
5. The subscriber may receive at most `n` more items. Back-pressure protects the consumer (and memory) from a faster producer: items are buffered up to a limit, and then the publisher must wait or drop.
6. When a subscriber's buffer is full and it has no outstanding demand. With `Runnable::run` the publishing thread is also the delivering thread, so blocking it can never be released — `offer` with a drop handler never blocks.
7. A mediator knows its colleagues and contains the coordination rules (it answers and routes); an event bus only broadcasts typed events and knows no rules. A facade simplifies access to a subsystem in one direction; the subsystem does not know the facade.
8. In: `logging` → `errorBoundary` → `authentication`; out: `authentication` → `errorBoundary` → `logging`. Yes — `authentication` returns the 401 through the outer links, so `logging` records it.
9. Only the originator may read its state, so the caretaker cannot depend on or corrupt it. A record memento gives up that opacity but is immutable, comparable with `equals`, safely shareable and validated in one place.
10. Memento: no inverse operation to write, restoring is always correct. Command: stores small operations instead of whole states, so it uses far less memory for large documents.

10 Assignments

- [01 — Live auction notifications](#) ★★☆☆ (Observer)
- [02 — Expense approval chain](#) ★★☆☆ (Chain of Responsibility)

11 Further reading

- Gamma, Helm, Johnson, Vlissides, *Design Patterns* (1994) — Observer, Mediator, Chain of Responsibility, Memento.
- [Reactive Streams specification](#) and the `java.util.concurrent.Flow` Javadoc.
- JEP 431 — [Sequenced Collections](#) · JEP 440 — [Record Patterns](#) · JEP 456 — [Unnamed Variables & Patterns](#)
- Martin Fowler, "Event Collaboration" and "Domain Event" — the ideas behind the capstone's event bus.
- Terms in Turkish: [docs/glossary.md](#)